

Mark Ibrahim

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EXPERIENCE

New York, NY

Staff AI Researcher, Fundamental AI Research (FAIR) at Meta Superintelligence Labs

July 2019 – Present

- Directing foundational research to measure and advance generalization in multimodal LLMs and AI agents
 - Published 30+ research papers at top AI conferences, earning 3 spotlight awards & 2 orals (top 1% at ICML & ICLR)
 - Led a team of 10 researchers to develop [OpenApps](#) (ICLR Oral), a research environment for computer-use agents
- Co-author of [Self-Supervised Learning Cookbook](#) w/ Yann LeCun + R. Balestriero, sets foundational recipes for SSL
- Architected a distributed GPU training pipeline in PyTorch, enabling multimodal training on 12 billion data samples
- Directed research for 13+ researchers (including 2 AI Residents, 6 Research Interns, and 4 visiting PhD Researchers)
- Instructor for Georgia Tech's Deep Learning course (+10k students); co-organized [ICML tutorial](#) with 400+ attendees

Senior Machine Learning Engineer, Center for Machine Learning at Capital One

Sep 2016 – June 2019

- Led Explainable AI team to build tools and research for explaining black-box deep learning models
 - Built open-source Python [library](#) to generate global explanations for neural network predictions
 - Published 2 interpretability research papers (NeurIPS workshop 2018 and ACM AAAI 2019) + Patent 16/855,685
- Engineered a real-time [notification](#) system for predicting mistaken charges on 10 million transactions per day
- Developed deep learning (RNN + LDA) customer archetype model in collaboration with Columbia Prof. John Paisley

Data Engineering Fellow (2016) & Technical Advisor, Insight Data Science

May 2016 – June 2018

- Developed a graph-based [knowledge search](#) engine parsing 5 million Wikipedia articles using Spark
- Advised incoming fellows on capstone projects, including problem selection and technical design

Quantitative Portfolio Risk Analyst, UBS

Jun 2012 – Aug 2014

- Applied unsupervised machine learning (PCA) to identify \$570k in uncaptured sensitivity to a 0.01% move in rates
- Automated daily 2½ hour manual risk calculation for \$658 million trading portfolio in Python

SELECT RESEARCH

Advancing Multimodal LLM Training and Evaluation

- Multimodal training objective with graph capturing [cross-sample](#) similarities w/ Yann LeCun (*ICLR*)
- [LLIP](#) a state-of-the-art open-weight vision-encoder (ViT-G) with optimized visual cross attention (*ICML*)
- [Common-O](#) multimodal reasoning (*NeurIPS*, senior author), [UniBench](#) (*NeurIPS*, senior author), [Battle of backbones](#) (*NeurIPS*)

Alignment for LLMs

- [Multi-token any-to-any](#) prediction for efficient knowledge encoding and [maze navigation](#) (*NeurIPS*, senior author)
- [AbstentionBench](#) (*NeurIPS*, co-lead) measuring LLMs' ability to say 'I don't know'; adopted in [GPT-5's system card](#)

Self-Supervised Learning (SSL) & Generalization

- [Occam's Razor](#): a minimalist SSL method (*TMLR*), [SSL CookBook](#) w/ Yann LeCun, [ImageNet-X](#) (*ICLR spotlight*, senior author)
- [Whac-A Mole Shortcuts](#) (*CVPR*, senior author), [Generalization across Geographies](#) of vision systems (*ICLR*, senior author)

Computer-use Agentic Systems

- Led team of 10 researchers to develop [OpenApps](#), a research environment for computer-use agents (*ICLR 2026 Oral*)
- Directing research into post-training approaches for reliable computer use agents trained with world models

EDUCATION

Statistics MicroMasters, Massachusetts Institute of Technology (MIT)

M.S. Mathematics, University of Vermont

Course Instructor: [Calculus I](#) (72 students) and [Calculus II](#) (38 students)

Honors B.A. Mathematics, Magna Cum Laude ([19th Gold Scholar](#)), Hamilton College